

SafeSignal Technologies Overview

This is a high-level summary of the main technologies used across the project.

Overall Stack

Area	Technologies
Mobile app	Expo, React Native, React, JavaScript
Moderator dashboard	React, Vite, Tailwind CSS, JavaScript
Backend API	Node.js, Express, Socket.IO
Database	PostgreSQL, PostGIS
ML service	Python, FastAPI, Uvicorn
ML models/providers	Google Gemini
Cache	Redis
Authentication	JWT, bcrypt, Google Sign-In
Maps	Google Maps, React Native Maps, deck.gl
Notifications	Socket.IO realtime events, Firebase Cloud Messaging, Expo Notifications, Notifee
Media storage	Local upload storage, optional Cloudflare R2 via S3-compatible API
Monitoring/logging	Sentry, Winston
Deployment	Render, Neon, Docker
Testing	Jest, Supertest, Node test runner, pytest-style ML tests

By Component

Mobile App

The mobile app is built with Expo and React Native. It uses React Navigation for screens, React Query for API data, React Native Maps for maps, Expo modules for location/media/notifications, and Secure Store or AsyncStorage for local persistence.

Moderator Dashboard

The dashboard is a React web app built with Vite and styled with Tailwind CSS. It uses React Router for navigation, React Query for API data, Google Maps and deck.gl for map/heatmap views, Lucide icons, and Socket.IO for realtime updates.

Backend

The backend is a Node.js and Express API. It uses PostgreSQL through pg-promise, JWT authentication, bcrypt password hashing, Socket.IO realtime events, Multer for uploads, Firebase Admin for push notifications, and Sentry/Winston for monitoring and logging.

Database

The database is PostgreSQL with PostGIS for geospatial incident data and location queries. It stores users, incidents, reports, moderation history, ML outputs, notifications, and witness-corroboration data.

ML Service

The ML service is a Python FastAPI service. a Gemini-backed provider for LLM analysis, embeddings, insights, and media judgment.

Deployment and Tooling

The deployment docs target Render for backend and ML hosting, Neon for PostgreSQL, Docker for the ML service, and Redis for caching. JavaScript dependencies are managed with npm.